

MARIZFIRST

Implementation Plan Document

Production-Level Healthcare Appointment Discovery + Assisted Booking Platform

Implementation Decision

Build MarizFirst as a production MVP in 75 working days for a solo senior full-stack developer with part-time UI/UX support. With 2 developers + dedicated UI/UX support, the same scope can be completed in 55-60 working days. This plan assumes Next.js, TypeScript, PostgreSQL, Prisma, custom RBAC admin, partner dashboard, official WhatsApp API, Cloudinary/S3 media storage, SEO architecture, analytics, QA, staging, and production launch.

Item	Plan
Product	MarizFirst - healthcare appointment discovery and assisted booking platform
Implementation target	Production MVP, not static website and not throwaway prototype
Core model	Hybrid booking: patient selects preferred slot; appointment stays Pending Confirmation until admin/partner approval
Main modules	Public website, booking system, admin panel, partner dashboard, notifications, SEO, analytics, security
Recommended timeline	75 working days / 15 working weeks for one strong full-stack developer + UI/UX support
Faster team option	55-60 working days with 2 developers + UI/UX designer + QA support
Launch approach	One focused local market first, then city-wise and speciality-wise expansion
Important rule	No discount/coupon positioning. Build trust, verified profiles, clarity and assisted confirmation.

Prepared for: Product planning, development, QA, deployment and launch execution

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1. Executive Implementation Strategy

MarizFirst should be implemented as a production-grade healthcare appointment assistance platform. The build cannot be treated like a simple doctor directory because the core business value depends on verified profiles, patient trust, slot-based appointment requests, admin/partner confirmation, WhatsApp updates, SEO discovery, and secure patient data handling.

The implementation must follow a modular architecture where public SEO pages, booking logic, admin operations, partner dashboard, media handling, notifications, audit logs, analytics, and security controls are built as connected parts of one system.

Non-Negotiable Product Rule

The website must not auto-confirm appointments in Phase 1. Every patient booking request starts as Pending Confirmation and becomes confirmed only after admin or healthcare partner action. This protects trust because clinics and doctors may change availability manually.

Strategy Area	Implementation Direction
Primary objective	Launch a reliable appointment request funnel with verified doctors, hospitals and clinics.
Primary conversion	Search/profile visit -> slot selection -> appointment request -> WhatsApp/admin/partner confirmation.
Engineering priority	Correct workflows, secure data access, stable database model, and operational admin control before visual polish.
Design priority	Dark premium healthcare UI, cyan glowing CTAs, glassmorphism cards, readable patient-first UX.
SEO priority	Server-rendered doctor, hospital, clinic, city, speciality and blog pages with metadata, schema and sitemap.
Security priority	RBAC, ownership checks, sensitive patient data protection, audit logs, rate limiting and secure deployment.

2. Team, Timeline and Delivery Model

The realistic production timeline depends on team size. The safest estimate for a production-level MVP with public website, admin panel, partner dashboard, booking engine, WhatsApp, SEO, analytics, security and QA is 75 working days with one strong full-stack developer and part-time UI/UX support.

Team Setup	Timeline	Notes
Solo senior full-stack + part-time UI/UX	75 working days / 15 weeks	Recommended if budget is controlled. Requires strict scope control and no extra features.
2 full-stack developers + UI/UX designer	55-60 working days / 11-12 weeks	Best balance for speed and quality. One developer owns public/SEO, second owns backend/admin/partner.
3-4 person product team	45-50 working days / 9-10 weeks	Faster, but requires strong product management and QA discipline.

Team Setup	Timeline	Notes
Static website only	15-25 working days	Not recommended. It will fail the appointment/admin/partner workflow requirements.

2.1 Recommended Core Team Responsibilities

Role	Responsibility
Product Owner / Business	Approve scope, launch city, specialties, partner onboarding rules, monetization, operational SOP.
UI/UX Designer	Dark premium UI system, homepage, listing pages, profile pages, booking flow, admin/partner UX, responsive rules.
Full-Stack Developer	Next.js app, database, APIs, admin dashboard, partner dashboard, booking engine, notifications, SEO.
QA / Tester	Functional, responsive, accessibility, security, SEO, and operational testing before launch.
Content / SEO	Doctor/hospital/clinic content, city/speciality content, blog setup, metadata, FAQs and legal copy coordination.
Operations Admin	Test real booking workflow, partner confirmation workflow, WhatsApp templates and launch SOP.

3. Final Scope for Production MVP

The MVP must be built around functional appointment operations. Visual design matters, but the product fails if admin cannot manage appointments, partners cannot respond, patient data leaks, SEO pages are weak, or notifications fail silently.

Module	Scope	Priority
Public Website	Home, doctors, hospitals, clinics, specialties, city pages, blog, contact, partner-with-us, legal pages.	P0
Search + Discovery	Search by doctor, hospital, clinic, speciality, city and area; filter listing pages; mobile filter drawer.	P0
Doctor Profiles	Profile image, professional details, speciality, city, location, fee, slots, FAQ, SEO metadata, schema, CTA.	P0
Hospital/Clinic Profiles	Facilities, gallery, linked doctors/team, address, map link, CTA, SEO metadata and schema.	P0
Booking Engine	Slot selection, short patient form, consent, Pending Confirmation status, source tracking, lead creation.	P0
Admin Panel	Manage listings, appointments, partners, leads, content, SEO, media, notifications, audit logs.	P0
Partner Dashboard	Own profile, slots, appointment accept/reject/reschedule, completion/no-show update, basic reports.	P0
WhatsApp Notifications	Acknowledgement, admin alert, partner alert, confirmation, reschedule, cancellation, reminder, logs.	P0
SEO Architecture	Dynamic metadata, sitemap, robots, schema, canonical URLs, internal linking, redirects.	P0
Security + Privacy	RBAC, ownership checks, rate limits, CAPTCHA, secure cookies, audit logs, backups, legal pages.	P0
Analytics	Search, profile view, slot selected, form started, appointment submitted, WhatsApp clicked, confirmed/cancelled.	P0

3.1 Explicitly Out of Scope for Phase 1

- No mobile app before the web funnel is proven.
- No full EMR/EHR system.
- No AI diagnosis, medical advice chatbot, treatment guarantee or emergency-care claims.
- No public reviews/ratings unless verified completed appointments are available.
- No online payment gateway until partner settlement, refund and cancellation rules are finalized.

- No teleconsultation/video consultation unless partner operations are ready.
- No unofficial WhatsApp Web automation. Use official WhatsApp Business API or manual fallback.

4. Work Breakdown Structure

Phase	Name	Deliverables
Phase 0	Preparation	Finalize scope, open decisions, launch city, first specialities, partner data fields, documents, design direction.
Phase 1	UX + Design System	Dark premium UI system, wireframes, high-fidelity screens, responsive behavior, component inventory.
Phase 2	Engineering Foundation	Repo, Next.js, TypeScript, Tailwind, Prisma, PostgreSQL, auth, env validation, CI.
Phase 3	Data Model + Backend Core	Schema, migrations, seed data, repositories, services, validation, RBAC, audit logs.
Phase 4	Public Website	Home, search, listings, profiles, static pages, blog pages, media rendering, SEO base.
Phase 5	Booking Engine	Slots, booking form, appointment creation, lead creation, status model, source tracking.
Phase 6	Admin Panel	Doctor/hospital/clinic CRUD, appointments inbox, partners, slots, content, SEO, notifications, audit logs.
Phase 7	Partner Dashboard	Profile management, slot management, appointment actions, scoped access and reports.
Phase 8	Notifications + Analytics	WhatsApp templates, queue, webhook, retries, GA4 events, server-side funnel events.
Phase 9	QA + Hardening	Functional, responsive, accessibility, security, SEO, performance, operational QA.
Phase 10	Launch + Monitoring	Production deployment, domain, SSL, Search Console, backups, monitoring, first-week operations.

5. Technical Setup and Engineering Standards

Area	Standard
Frontend	Next.js App Router, TypeScript, server-rendered SEO pages, client components only for interactivity.
Styling	Tailwind CSS, design tokens, dark premium theme, responsive layouts, accessible focus states.
Database	PostgreSQL with Prisma migrations. No media binary storage inside DB.
Media	Cloudinary for MVP or S3-compatible storage. DB stores URL/publicId/metadata only.
Validation	Zod schemas reused server-side; React Hook Form client-side.
Auth	NextAuth/Auth.js or controlled credentials/OTP. Secure HTTP-only session cookies.
RBAC	Permission guard on every dashboard route and API; partner ownership filtering at query level.
Services	Business logic inside service modules; route handlers should stay thin.
Repositories	Database access in repositories/query modules, not scattered in UI components.
Logging	Structured logs; do not log patient concern text to public analytics or unsafe logs.
Testing	Unit, integration, E2E, responsive, accessibility, SEO, security and performance testing.
Deployment	Vercel + managed PostgreSQL + Cloudinary/S3 + Sentry + uptime monitoring.

6. Day-by-Day Production Implementation Plan

The following plan is designed for 75 working days. Every day has a clear objective, execution steps and exit criteria. This is a production MVP plan, not a learning-only checklist.

Phase 0 - Preparation

Day	Focus	What to Do	Deliverable	Exit Criteria
Day 1	Kickoff + scope lock	Review PRD/TRD/UI brief/backend schema/app flow. Freeze P0 scope. Confirm no discount/coupon positioning. Define single launch market.	Approved MVP scope and decision log.	No P0 ambiguity remains for public site, booking, admin, partner, SEO, notifications.
Day 2	Launch data model decisions	Finalize launch city, first specialties, doctor/hospital/clinic onboarding fields, verification checklist, fee visibility rules, appointment SLA.	Partner data collection template.	Doctor listing fields and hospital/clinic profile fields approved.
Day 3	User journeys + operational SOP	Map patient, admin, partner, content editor and notification journeys. Write operational SOP for Pending Confirmation.	Workflow maps and SOP draft.	Every appointment status has owner, trigger and next action.
Day 4	Information architecture	Finalize sitemap, URL patterns, city/specialty SEO structure, admin menu, partner menu, legal pages.	Final IA and route list.	All routes mapped to owner, content source and SEO priority.
Day 5	Backlog + acceptance criteria	Create implementation backlog with P0/P1/P2, acceptance criteria, Definition of Done and risk list.	Sprint-ready backlog.	Developer can start without guessing scope.

Phase 1 - UX/UI

Day	Focus	What to Do	Deliverable	Exit Criteria
Day 6	Design system foundation	Define dark premium healthcare theme: colors, type scale, spacing, radii, shadows, glass cards, button states, status tokens.	Design tokens and component mood board.	Buttons and cards match approved reference direction.
Day 7	Homepage wireframe + UI	Create desktop and mobile homepage structure: hero, search, trust strip, specialties, doctors, hospitals, clinics, how it works, footer CTA.	Homepage UI v1.	Hero CTA and search are above the fold and mobile-friendly.
Day 8	Listing UX	Design doctors/hospitals/clinics listing pages with filters, mobile drawer, empty states and card hierarchy.	Listing screens.	Users can compare important data without opening every profile.
Day 9	Profile UX	Design doctor, hospital and clinic profile pages with sticky mobile CTA, slots, details, FAQ, privacy/trust notes.	Profile screens.	Profile page supports conversion without forcing login.
Day 10	Booking flow UX	Design slot picker, patient form, consent, pending confirmation screen, error states and success states.	Booking flow screens.	Form is short, accessible and clear about Pending Confirmation.
Day 11	Admin dashboard UX	Design admin dashboard, doctor wizard, appointment inbox, status actions, media upload, SEO/FAQ tabs, audit timeline.	Admin screens.	Admin can list a doctor using step-by-step wizard.
Day 12	Partner dashboard UX	Design partner dashboard: own profile, slots, appointment requests, reschedule/reject/accept, reports.	Partner screens.	Partner cannot see global data in UX or backend plan.

Phase 2 - Engineering Foundation

Day	Focus	What to Do	Deliverable	Exit Criteria
Day 13	Project setup	Create private GitHub repo, Next.js App Router, TypeScript, Tailwind, ESLint, Prettier, folder structure.	Running app skeleton.	npm run lint, typecheck and build pass.
Day 14	Theme implementation	Implement Tailwind theme tokens, global layout, dark premium shell, base UI components: Button, Card, Badge, Input, Modal, Drawer.	UI foundation components.	Components are reusable and responsive.
Day 15	Database setup	Set up PostgreSQL, Prisma,	DB connected locally.	Prisma generate/migrate/seed

Day	Focus	What to Do	Deliverable	Exit Criteria
		env validation, migration workflow, seed structure.		works.
Day 16	Auth foundation	Implement auth provider, sessions, roles enum, protected route middleware, login pages.	Auth base.	Admin and partner routes require login.
Day 17	CI + code quality	Set up GitHub actions: install, lint, typecheck, test placeholder, build. Create README setup guide.	CI pipeline.	Pull request cannot merge if build fails.
Day 18	Core layouts	Implement public layout, admin layout, partner layout, navigation, footer, protected dashboard sidebar.	Layout shell.	All route groups render correctly.

Phase 3 - Backend Core

Day	Focus	What to Do	Deliverable	Exit Criteria
Day 19	Core schema - users/location	Implement User, Role, City, Location, Speciality models and seed data.	Prisma migration 1.	Cities and specialities can be seeded and queried.
Day 20	Profile schema	Implement Doctor, Hospital, Clinic, relations, status fields, SEO fields, verification fields.	Profile data models.	Doctor can link to city/speciality and hospital/clinic.
Day 21	Media schema + upload base	Implement MediaAsset model and Cloudinary/S3 upload service with validation.	Media upload service.	Image uploaded externally and metadata stored in DB.
Day 22	Slots + appointments schema	Implement AppointmentSlot, Patient, Appointment, Lead models and status enums.	Booking data models.	Appointments support Pending Confirmation and source tracking.
Day 23	Notifications + audit schema	Implement Notification, AuditLog, FAQ, BlogPost, Redirect, ListingPlan/Commission fields.	Operational models.	Critical state changes have audit destination.
Day 24	Repositories	Create repositories for doctor, hospital, clinic, slot, appointment, patient, lead, media, notification, audit.	Repository layer.	Route handlers do not call Prisma directly for domain actions.
Day 25	Service layer	Implement DoctorService, AppointmentService, SlotService, NotificationService, AuditLogService, SeoService.	Service layer.	Business rules are testable outside UI.
Day 26	RBAC + permission guards	Implement role permissions, ownership checks, partner-scoped query helpers, export restrictions.	Permission system.	Partner cannot access another partner record via API.

Phase 4 - Public Website

Day	Focus	What to Do	Deliverable	Exit Criteria
Day 27	Homepage build	Implement dark premium homepage hero, search module, trust strip, specialities, featured profiles, CTA/footer.	Homepage v1.	Responsive at 360/390/430/1366/1440/1920.
Day 28	Global search API/UI	Build /api/search, autocomplete, result type labels, safe sanitization and empty state.	Search working.	Search returns doctors/hospitals/clinics/specialities/cities.
Day 29	Doctors listing	Build /doctors with filters, cards, pagination/cursor loading and mobile filter drawer.	Doctor listing page.	Listing shows verified profile, speciality, location, fee, availability and CTA.
Day 30	Doctor city/speciality pages	Build /doctors/[city] and /doctors/[speciality]/[city] with SEO metadata and canonical logic.	SEO doctor landing pages.	Pages are indexable and server-rendered.
Day 31	Doctor profile page	Build /doctor/[slug] with profile details, media, locations, slots, FAQ, trust notes, sticky mobile CTA.	Doctor profile page.	Patient can select preferred slot from profile.
Day 32	Hospital listing/profile	Build /hospitals and /hospital/[slug], linked doctors, facilities, gallery and CTA.	Hospital pages.	Hospital profile shows linked doctors and lead action.
Day 33	Clinic listing/profile	Build /clinics and /clinic/[slug], doctors/team, facilities, gallery, slots/CTA.	Clinic pages.	Clinic can convert to appointment request.
Day 34	Static/legal/support pages	Build contact, partner-with-us,	Static pages.	Legal pages are accessible

Day	Focus	What to Do	Deliverable	Exit Criteria
		privacy policy, terms, medical disclaimer, partner terms, data deletion.		from footer.
Day 35	Blog pages	Build blog listing/detail, categories, author, thumbnail, metadata and rich content rendering.	Blog module public side.	Published blogs appear publicly and are SEO-ready.
Day 36	Public polish pass	Fix visual spacing, loading states, empty states, responsive issues and image optimization.	Public website beta.	No broken public layout at required breakpoints.

Phase 5 - Booking Engine

Day	Focus	What to Do	Deliverable	Exit Criteria
Day 37	Slot picker component	Implement available slot display, date grouping, past slot exclusion, capacity visibility and loading states.	Slot picker.	Only selectable slots are available and not expired.
Day 38	Booking form UI	Implement short form: name, phone, city, optional email, concern, consent. Add validation and accessible errors.	Booking form.	Invalid phone/consent does not submit.
Day 39	Appointment creation API	Build /api/appointments POST with transaction: validate, patient upsert, slot check, appointment create, lead create, audit log.	Booking API.	Appointment always starts Pending Confirmation.
Day 40	Source tracking	Capture source page, profile type, UTM fields, referrer and analytics event.	Lead attribution.	Admin can see where appointment came from.
Day 41	Booking confirmation screen	Show request received, Pending Confirmation, next steps, support CTA and safe user-facing copy.	Confirmation UI.	Patient never sees fake confirmed state.
Day 42	Duplicate/spam protection	Add rate limiting, CAPTCHA/Turnstile hook, duplicate recent request guard, safe error messages.	Spam protection.	Repeated submissions are throttled/challenged.
Day 43	Status transition logic	Implement allowed transitions: pending, confirmed, rescheduled, cancelled, completed, no-show, lost.	Status service.	Invalid transitions fail safely.
Day 44	Booking integration tests	Test search -> profile -> slot -> form -> pending appointment -> lead -> audit log.	Booking tests.	Core booking scenario passes.

Phase 6 - Admin Panel

Day	Focus	What to Do	Deliverable	Exit Criteria
Day 45	Admin dashboard overview	Build KPI cards: pending, confirmed, cancelled, completed, no-show, leads, SLA warnings, notification failures.	Admin dashboard.	Admin can see operational status.
Day 46	Doctor listing wizard - basic/professional	Build Add Doctor wizard steps 1-2: basic and professional details with validation.	Doctor wizard start.	Incomplete required fields cannot publish.
Day 47	Doctor listing wizard - location/association	Build steps 3-4: city, area, hospital/clinic association, fee and availability rules.	Doctor location/availability wizard.	Doctor can link to hospital/clinic and create slot rules.
Day 48	Doctor listing wizard - media/SEO/FAQ	Build steps 5-7: media upload, SEO fields, FAQ, verification checklist, preview.	Doctor wizard complete.	Admin can publish SEO-ready doctor profile.
Day 49	Hospital/clinic CRUD	Build hospital and clinic create/edit with gallery, facilities, linked doctors/team, location, SEO and status.	Hospital/clinic admin.	Hospital and clinic profiles can be published.
Day 50	Appointments inbox	Build appointment table with filters by status, date, doctor/hospital/clinic, city, source and SLA.	Appointment inbox.	Admin can find pending requests quickly.
Day 51	Admin status actions	Build confirm, reschedule, cancel, complete, no-show, lost actions with notes and reason handling.	Status actions.	Each status action updates appointment and writes audit log.
Day 52	Slots admin	Build slot view, block slots,	Admin slot manager.	Admin can fix partner slot

Day	Focus	What to Do	Deliverable	Exit Criteria
		bulk creation, conflict detection and capacity override rules.		issues manually.
Day 53	Partners and roles	Build partner account creation, role assignment, status, organization linking, permissions and plan fields.	Partner admin.	Partners can be created and scoped correctly.
Day 54	Content + SEO admin	Build blog CRUD, FAQ management, metadata fields, redirects and sitemap inclusion controls.	Content/SEO admin.	Content editor can publish without patient data access.
Day 55	Notifications + audit admin	Build notification log, retry action, audit log search/filter and timeline display.	Operational visibility.	Failed notification is visible and retryable.
Day 56	Admin QA pass	Test doctor publish, hospital publish, appointment confirm/reschedule/cancel, audit logs and role restrictions.	Admin beta.	Admin can operate core business without developer help.

Phase 7 - Partner Dashboard

Day	Focus	What to Do	Deliverable	Exit Criteria
Day 57	Partner dashboard shell	Build partner login redirect, dashboard home, scoped navigation, own profile summary.	Partner shell.	Partner cannot access admin routes.
Day 58	Partner profile management	Allow partner to edit own profile fields with optional admin-review flag for sensitive fields.	Partner profile module.	Partner cannot edit unrelated profile.
Day 59	Partner slot management	Build create/edit/block/bulk slot UI with conflict prevention and past-slot guard.	Partner slots.	Partner can manage availability safely.
Day 60	Partner appointment actions	Build accept, reject, request reschedule, mark completed/no-show with notes.	Partner appointments.	Partner actions respect allowed transitions.
Day 61	Partner reports/support	Build basic reports and support request flow to admin.	Partner reports.	Partner can view own appointment summary only.
Day 62	Partner security QA	Test URL manipulation, API ownership checks, content editor restrictions and partner export restrictions.	Partner security report.	No cross-partner data access possible.

Phase 8 - Notifications/SEO/Analytics

Day	Focus	What to Do	Deliverable	Exit Criteria
Day 63	WhatsApp provider setup	Select provider, configure templates, env secrets, webhook URL, sandbox/test numbers.	WhatsApp setup.	Template approval process started or sandbox working.
Day 64	Notification queue	Build notification create/send/retry logic for acknowledgement, admin alert, partner alert, confirmation, reschedule, cancel, reminder.	Notification engine.	Appointment creation does not fail silently if provider fails.
Day 65	Webhook handling	Build /api/webhooks/whatsapp with signature verification, idempotency, providerMessageId updates and failure logging.	Webhook integration.	Delivery statuses update notification records.
Day 66	SEO technical completion	Implement generateMetadata, JSON-LD schema, sitemap, robots, canonical, internal links, image alt rules, redirects.	SEO architecture.	Published profiles and blogs appear in sitemap.
Day 67	Analytics implementation	Add GA4 events and server-side operational events without leaking phone/email/concern.	Analytics events.	Search, profile, slot, booking and status events are tracked.

Phase 9/10 - QA + Launch

Day	Focus	What to Do	Deliverable	Exit Criteria
Day 68	Functional QA	Run full patient, admin, partner, content and	Functional QA report.	All P0 workflows pass.

Day	Focus	What to Do	Deliverable	Exit Criteria
		notification workflows. Fix blocking bugs.		
Day 69	Responsive + UI QA	Test mobile 360/390/430, tablet, desktop 1366/1440/1920; fix card, drawer, table and sticky CTA issues.	Responsive QA report.	No horizontal scroll or clipped CTAs.
Day 70	Accessibility QA	Test labels, keyboard flow, focus states, contrast, drawers/modals, errors and tap targets.	Accessibility fixes.	Booking flow usable without mouse.
Day 71	Security QA	Test RBAC bypass, rate limits, input validation, webhook secret, secure cookies, export logs and partner ownership.	Security QA report.	Sensitive data is not exposed.
Day 72	Performance + SEO QA	Optimize images, Core Web Vitals, listing pagination, metadata, schema, sitemap, robots and canonical.	Performance/SEO report.	Public pages are fast and indexable.
Day 73	Staging UAT	Deploy staging, import realistic test data, run admin/partner UAT and WhatsApp fallback process.	UAT signoff.	Operations team can handle appointments.
Day 74	Production deployment	Configure production env, domain, SSL, DB backups, monitoring, Sentry, Search Console, GA4, seed real launch data.	Production build.	Live site passes smoke test.
Day 75	Launch + first-week plan	Run launch checklist, monitor errors, pending appointments, WhatsApp failures, speed, indexing and user feedback.	Launch report and post-launch backlog.	MVP is live with monitoring and support SOP.

7. Detailed Module Implementation Notes

7.1 Doctor Listing Implementation

Doctor listing must be built as a step-by-step admin wizard. A single long form will create incomplete profiles and poor SEO quality. The doctor listing flow must include profile, verification, location, speciality, media, slots, SEO, FAQ and publish status.

Wizard Step	Fields / Actions	Rules
1. Basic Details	Name, slug, gender, profile image, phone, email, languages, short bio, status.	Phone/email are internal; public page should not expose private contact unless approved.
2. Professional Details	Speciality, sub-speciality, qualification, experience, registration number, council, designation, expertise.	Registration number improves trust; mark optional only if not available.
3. Location & Association	City, area, address, map URL, consultation location, linked hospital/clinic.	Doctor may work at multiple locations later; MVP can start with primary location.
4. Fee & Availability	Consultation fee, follow-up fee, fee visibility, consultation days, time range, slot duration, capacity.	No online payment in Phase 1. Fee is informational.
5. Media Upload	Profile photo, clinic image, gallery/certificate optional, alt text.	Upload to Cloudinary/S3; store only metadata in DB.
6. SEO & FAQ	SEO title, meta description, canonical, FAQ, schema fields.	Doctor page must be indexable and sitemap-ready after publish.
7. Verification	Checklist: name, qualification, registration, location, fee, media, no fake reviews, no medical guarantees.	Cannot publish if required verification fields fail.
8. Preview & Publish	Preview final public profile and publish as Active.	Publishing writes audit log.

7.2 Media Storage Implementation

- Do not store image/video binaries inside PostgreSQL or MongoDB for this project.
- Use Cloudinary for MVP or AWS S3 + CloudFront for larger scale.
- Store media URL, publicId, altText, fileName, mimeType, width, height, sizeBytes, ownerType and usageType in MediaAsset table.
- Require alt text for SEO-important images such as doctor profile, hospital cover, clinic cover and blog thumbnail.
- Convert images to WebP/AVIF where possible and serve through CDN.
- Use folder naming like marizfirst/doctors/doctor-slug/profile.webp and marizfirst/blogs/blog-slug/thumbnail.webp.

7.3 Booking Engine Implementation Rules

- Patient can select a visible preferred slot, but appointment status starts as PENDING_CONFIRMATION.
- Appointment creation should validate phone, city, selected slot, consent, rate limit and slot capacity.
- Patient creation/upsert, appointment creation, lead creation and audit log should run in a transaction where practical.
- External WhatsApp failure should not silently lose the appointment. Create notification record and retry/failure visibility.
- Allowed appointment statuses: PENDING_CONFIRMATION, CONFIRMED, RESCHEDULED, CANCELLED, COMPLETED, NO_SHOW, LOST.
- Every status change requires audit log with actor, timestamp, old value, new value and reason/notes where needed.

8. Admin Panel Build Plan

Admin Module	Implementation Requirements	Priority
Dashboard	Operational overview with pending SLA, appointment counts, lead sources, top partners, notification failures.	P0
Doctors	Step-by-step doctor listing wizard, edit, verify, publish/unpublish, media, SEO, FAQ, slots.	P0
Hospitals/Clinics	CRUD, galleries, facilities, linked doctors/team, SEO, status, location and CTA settings.	P0
Appointments	Inbox, filters, confirm, reschedule, cancel, complete, no-show, lost, notes and notification status.	P0
Slots	View all partner slots, block unavailable dates, bulk creation, capacity management and conflict resolution.	P0
Partners	Create accounts, roles, organization linking, plans, commission fields and status.	P0
Leads	Source tracking, lead status, notes, export with permission gate and audit logging.	P0
Content	Blogs, FAQs, city/speciality content, homepage content and static/legal pages.	P0
SEO	Metadata, slugs, canonical, schema, redirects and sitemap inclusion.	P0
Audit Logs	Track sensitive changes, exports, status changes, login failures and permission changes.	P0

9. Partner Dashboard Build Plan

Partner Module	Implementation Requirements	Priority
Login + Scope	Partner logs in and sees only assigned profile, slots and appointment requests.	P0
Profile Management	Partner edits own non-sensitive fields; sensitive edits can require admin approval.	P0
Slot Management	Add/edit/block/bulk create slots; prevent past slots, conflicts and overcapacity.	P0
Appointment Requests	Accept, reject, request reschedule, mark completed/no-show with notes and reason.	P0
Patient Details	Show only appointment-specific patient data. No unrelated leads or global exports.	P0
Reports	Pending, confirmed, completed, no-show and source summary by date range.	P1
Support Requests	Partner can request profile/content/operational updates from admin.	P1

10. Public Website and SEO Build Plan

SEO Area	Implementation Requirement
Rendering	Doctor, hospital, clinic, city, speciality and blog pages should render server-side/indexable.
Metadata	Use generateMetadata for title, description, canonical and Open Graph where applicable.
Schema	Use Physician, MedicalOrganization, LocalBusiness, BreadcrumbList and FAQPage where appropriate.

SEO Area	Implementation Requirement
Sitemap	Generate sitemap dynamically from active published profiles, city pages, speciality pages and blog posts.
Robots	Allow public pages; block admin, partner dashboard, private APIs and staging URLs.
Internal Linking	Doctor profile links to speciality/city/hospital/clinic pages; city pages link to relevant profiles.
Image SEO	Descriptive filenames, alt text, optimized sizes and lazy loading below fold.
Content Quality	No medical guarantees, no fake urgency, no fake reviews, no discounted messaging.

11. Booking Engine and Notification Build Plan

Notification Type	Implementation Requirement
Acknowledgement	Patient receives request received + Pending Confirmation message after appointment submission.
Admin Alert	Admin receives new request with patient details, provider, source, selected slot and action link.
Partner Alert	Partner receives scoped appointment request with accept/reject/reschedule action.
Confirmation	Patient receives confirmed date/time, provider, address and support contact.
Reschedule	Patient and partner receive old slot, new slot and reason if available.
Cancellation	Patient and partner receive cancellation information and support option.
Reminder	Patient receives reminder 12-24 hours before confirmed appointment.
Failure Handling	Every notification stores providerMessageId, status, sentAt, errorMessage and retryCount.

12. QA, Security and Launch Readiness

QA Area	Checklist
Functional QA	Search -> profile -> slot -> booking -> admin confirm -> WhatsApp status; partner accept/reject; content publish.
Responsive QA	360, 390, 430, tablet, 1366, 1440 and 1920 widths. No horizontal scroll or clipped sticky CTA.
Accessibility QA	Labels, keyboard navigation, focus states, contrast, form errors, modals/drawers and tap targets.
Security QA	RBAC bypass, partner ownership, input validation, webhook secret, rate limits and export restrictions.
SEO QA	Metadata, schema, sitemap, robots, canonical, internal links, redirects and indexability.
Performance QA	Image sizes, Core Web Vitals, pagination, caching, bundle size and dashboard API loading states.
Operational QA	Admin can handle pending SLA, no partner response, reschedule/cancel, notification retry and fallback WhatsApp.

13. Deployment, Monitoring and Post-Launch Operations

Environment / Area	Requirement
Local	Developer env, seed data, .env.example, README setup, local DB or dev DB.
Staging	Production-like environment, protected access, test data, WhatsApp sandbox/templates, UAT workflows.
Production	Domain, SSL, production DB, secure secrets, Cloudinary/S3, backups, monitoring, analytics, Search Console.
CI/CD	Lint, typecheck, tests and build on every pull request; staging before production.
Backups	Daily DB backups and periodic restore test. Media backup/export plan where needed.
Monitoring	Sentry, uptime monitor, structured logs, notification failure dashboard and SLA alerts.
Rollback	Document rollback for frontend deploy and migration rollback/forward-fix strategy.

14. Risks, Dependencies and Open Decisions

Risk	Impact	Mitigation
Fake availability	Patient trust breaks if displayed slots are not real.	Use Pending Confirmation, SLA dashboard, partner escalation and admin manual override.
Slow partner response	Users may abandon or complain.	15-30 minute working-hour SLA, admin alerts, WhatsApp support path.
Low-quality profile data	Poor SEO, weak trust and low conversion.	Verification checklist before publish, required fields, content review.
WhatsApp approval delay	Notification launch risk.	Start template approval early; build manual fallback copy/action in admin.
Spam leads	Operational waste and partner dissatisfaction.	Rate limiting, CAPTCHA/Turnstile, duplicate guard, optional phone OTP later.
Partner data leakage	Privacy and legal risk.	RBAC, ownership-scoped queries, audit logs and security tests.
Overbuilding early	Timeline and cost increase.	Do not build mobile app, EMR, teleconsultation, payments or AI diagnosis in Phase 1.

14.1 Open Decisions Before Development Starts

- Final launch city and first 5-10 specialities.
- Exact doctor, hospital and clinic onboarding fields and verification process.
- Whether patient phone OTP is required at launch or only after spam appears.
- Custom admin vs Payload CMS final decision. Recommendation: custom admin for complex appointment operations.
- WhatsApp Business API provider and template approval timeline.
- Commission/listing plan fields and partner agreement terms.
- Which partner profile edits require admin approval.
- Initial content count: number of doctors, hospitals, clinics, city pages, speciality pages and blog posts for launch.

15. Final Delivery Checklist and Definition of Done

- All P0 public pages, admin modules, partner dashboard modules, booking flow, notifications, SEO and security requirements are implemented.
- Search -> profile -> slot -> booking -> Pending Confirmation -> admin/partner action -> patient notification works end-to-end.
- Doctor listing wizard includes basic details, professional details, location/association, fee/availability, media, SEO/FAQ, verification and preview/publish.
- No patient data is visible publicly or across unrelated partner accounts.
- All public pages are responsive, accessible, SEO-ready and fast on mobile.
- Admin can manage profiles, appointments, partners, leads, content, SEO, notifications and audit logs without developer help.
- Partner can manage only own profile, slots and appointment requests.
- WhatsApp notification failures are logged and retryable.
- Legal pages and medical disclaimer are live and linked in footer.
- GA4/Search Console/conversion events are configured without leaking sensitive patient data.
- Backups, monitoring, production secrets and rollback process are configured.
- Functional, responsive, accessibility, security, SEO, performance and operational QA checks pass.

Final Timeline Recommendation

Plan 75 working days for a production-level MVP with one strong full-stack developer and part-time UI/UX support. Do not compress below 60 working days unless there are at least two developers and the scope is frozen. The project can go live earlier only if admin/partner dashboard scope is reduced, but that would weaken the core operating model.